

Optimizing Oil Palm Detection: A Deep Dive into Algorithm Performance and Software Metrics

Jeklin Harefa

Computer Science Department
BINUS Graduate Program – Doctor of Computer Science Bina
Nusantara University
Jakarta, Indonesia
jeklin.harefa001@binus.ac.id

Yulyani Arifin

Computer Science Department
BINUS Graduate Program – Doctor of Computer Science Bina
Nusantara University
Jakarta, Indonesia
yulyaniarifin@binus.ac.id

Haryono Soeparno

Computer Science Department
BINUS Graduate Program – Doctor of Computer Science Bina
Nusantara University
Jakarta, Indonesia
haryono@binus.ac.id

Ford Lumban Gaol

Computer Science Department
BINUS Graduate Program – Doctor of Computer Science Bina
Nusantara University
Jakarta, Indonesia
fordlg@binus.ac.id

Abstract – Focus of this research is the detection of oil palm trees with aerial drone images by comparing several YOLO nano models, including: YOLOv10n, YOLOv11n, and YOLOv12n. The study conducted training and evaluated the YOLO version models with a different batch size to obtain some key detection metrics such as precision, recall, and mean Average Precision (mAP), with computational efficiency factors like inference latency and GPU memory consumption. The results show that all models can achieve accuracy with precision and recall values of more than 85% and mAP@50 around 89%. Meanwhile, YOLOv11n with a batch size of 64 was claimed as the fastest processing speed, which is around 15.70 ms per image, while YOLOv11n batch 32 achieved a promising result in mAP@50-95 (0.5671) with the lowest GPU allocation memory (41.97 MB). Based on the result, YOLOv11n has promising results in terms of speed, accuracy, and memory allocation so that it is very possible to be applied to drones that are able to detect oil palm trees in real time.

Keywords: oil palm trees, YOLO, monitoring, drone, images

I. INTRODUCTION

In global context, oil palm plantations have an important role in the vegetable oil industry. As one of the producer and suppliers of oil palm, Indonesia has a very significant contribution both on local and global scale[1]. With more than 16 million hectares of oil palm plantations spread across major regions including Riau, West Kalimantan, Central Kalimantan, North Sumatra, and Papua, Indonesia has become the top producer of Crude Palm Oil (CPO) in the global context.

The importance of monitoring oil palm trees is influenced by the effectiveness and efficiency of plantation monitoring. With proper monitoring methods and identification, oil palm plantations can achieve optimal yields. One of the traditional methods often used to monitor oil palm trees is by doing manual calculations, which is very prone to errors both in documentation and administrations, which may take a very long time, making it less efficient [2]. When the number of oil palm trees were very large and were counted one by one by some people, of course this could have a significant impact on next future of decision making.

Another way that can be used to detect and count the number of oil palm trees is by using images taken by a remote sensing done. The use of drones with high-resolution images makes counting oil palm trees more efficient and effective. Humans simply need to maneuver the drone and capture the tree images to accurately count the number of oil palm trees. Meanwhile, calculations carried out through manual field surveys or using GPS (Global Positioning System) to record locations require high costs and take a very long time [3]. With the existence of drone technology that utilizes computer vision, it can provide an impact in terms of ease in detecting oil palm trees automatically, speeding up tree counting, and increasing the accuracy and efficiency in oil palm plantation management

Currently, there are quite a few computer vision approaches that implement deep learning which are utilized to perform object detection. One of the algorithms that is widely used in real-time object detection is YOLO (You Only Look Once). Known as an algorithm that has fast and accurate detection, YOLO embedded into drones has the potential impact in real time for various fields, such as monitoring oil palm plantations. With YOLO's ability to detect more than 9000 objects in real time, this algorithm can be considered as one of the powerful and fast models in detecting and classifying object [4]. Starting from 2016, YOLO started to be known and had several improved versions. Until today, YOLO is still widely used in numerous fields, including medical image analysis [5], remote sensing object detection [6], smart transportation [7] and agricultural area [8], as an effective method for automatic object detection with high accuracy. Since YOLO (You Only Look Once) has progressed rapidly, and the latest versions (V10, V11, V12) provide considerable improvements both in performance and efficiency. However, the optimal implementation of this algorithm for oil palm detection in drone imagery, as well as the impact of varying batch sizes on training and inference, remains an underexplored area of research.

The key contribution of this research including:

1. A comprehensive comparative of the latest YOLO models (v10, v11, and v12) for oil palm tree detection with varying batch size of 8, 16, 32 and 64.

2. Balancing of detection quality and computational efficiency by analysing how batch size and model architecture affect to core software quality metric such as accuracy and the system performance.
3. Evaluate the effectiveness of YOLO model selections, which is visualized in metric values on test data.

This paper will be composed of several chapters: Chapter 2 will discuss research related to deep learning algorithms for oil palm plantation. Chapter 3 will discuss the proposed research workflow. Chapter 4 will discuss the experimental results and some discussions. Chapter 5 will provide information about the paper's summary and how it might be improved better.

II. LITERATURE REVIEW

Research on plantation object detection is progressing quickly, because of the increasing usage of drone technology and deep learning algorithms. In general, CNN (Convolutional Neural Network)-based models have significantly increased the detection of oil palm trees from drone images. Compared to many deep learning methods, YOLO (You Only Look Once) is the best model because it strikes the best balance between speed and accuracy. This model allows for monitoring and analyzes agricultural surroundings in real time, which is very important for larger applications like managing oil palm plantations [9].

Prasvita et al. [10] developed a methodology utilizing deep learning for identification of oil palm trees with YOLOv5. By using specific data from an oil palm region in Kalimantan, Indonesia, this study evaluates the performance of many YOLOv5 network models, including YOLOv5s (small), YOLOv5m (medium), YOLOv5l (large), and YOLOv5x (extra-large). This study including various processes: data acquisition, bounding box labelling, training and testing dataset splitting, model training, and evaluation. With a batch size of 32, the model achieved mAP50 of 0.851 (mean Average Precision at IoU = 0.50), mAP50-95 of 0.457 (a mean Average Precision at IoU = [0.50, 0.95]) and F1-Score of 0.785.

In 2024, Soetrisno et al. significantly advanced the creation of an ideal CNN model for oil palm detection by using EfficientDet-Lite2 [11]. Their investigation focused on adjusting the EfficientDet-Lite model's hyperparameters and input size, which led to better performance in the dumpy variety's ripeness classification. They enhanced the model by adding three box classes per layer to better handle the classification tasks and five layers of Bi-directional Feature Pyramid Networks (Bi-FPN), a method that focuses on capturing multi-scale information. Using the EfficientDet D2 model, this experiment yields an accuracy of 84%.

Similar with Soetrisno et al., In 2024, Minarto et al. developed a YOLOv5 algorithm in 2024 to count oil palms and use the Normalized Difference Vegetation Index (NDVI) to assess the plantation's normalcy [12]. The comprehensive RGB (Red Green Blue) images captured by an unmanned aerial vehicle (UAV) above a dense oil palm plantation in Indonesia's North Sumatra Province comprised the dataset used in their study. The F1-score increase from 0.81 to 0.83, indicating a slight improvement over previous study, while the confusion matrix showed an NDVI accuracy of 72.7%.

Naftali et al. introduced a novel method for estimating oil palm yield in the same year (2024) using a variety of YOLO models[13]. They used a dataset of fresh oil palm fruit that was collected on the plantation in Kalimantan, Indonesia. They examined how well the YOLOv8s Depthwise model performed in comparison to several YOLO models, such as YOLOv6s, YOLOv6l, YOLOv7 Tiny, YOLOv7l, YOLOv8s, and YOLOv8l. With $mAP@50 = 0.75$, $mAP@50-95 = 0.481$, an MAE (Mean Absolute Error) of 0.164, an RMSE (Root Mean Squared Error) of 0.4, and an inference time of around 0.027 seconds, the YOLOv8 Depthwise model performed better.

Furthermore, Haq et al. proposed modifying the core of Faster R-CNN (Region Based Convolutional Neural Network) to increase the effectiveness of oil palm tree detection utilizing UAV data [14]. They replaced the original backbone with EfficientNetV2-S, a lightweight and efficient deep learning architecture known for its high accuracy and computational efficiency. This backbone modification achieved the highest performance of evaluation results. The F1-Score for this model was 80% at IoU@50 and 73% at IoU@75, which shows that the model performed well at both object localization and classification tasks.

Heng et al. created a model that used drone images to classifying healthy and unhealthy oil palm trees in 2025 [15]. They explored several pre-trained models, including SSD (Single-Shot MultiBox Detector) and Faster R-CNN, with a variative of backbone modules. Their investigation led them to the conclusion that the Faster R-CNN ResNet101 FPN model was the best model for identifying and categorizing the health states of oil palm trees, outperforming all other models with AParea = all of 0.355, ARarea = all of 0.44, and total loss of 0.2296

III. METHODOLOGY

3. 1. Research Flow

This study consists of oil palm dataset collection from Roboflow, preprocessing to clean the data and using YOLO model to train the data, and evaluation to get the precision, recall, mAP@50, mAP@50-95, latency and GPU. Both of models will be fine-tuned to check the optimum score. The resulting models will be analyzed through classification metrics to evaluate the most effective model for detecting oil palms in drone images. The proposed research workflow can be seen on Fig. 1.

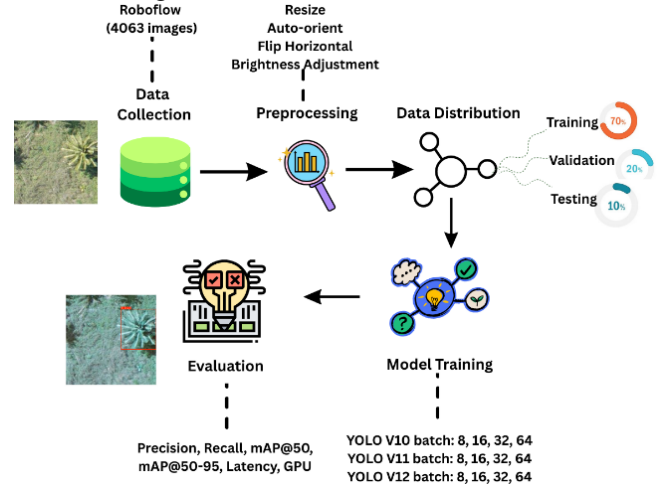


Fig. 1. The proposed research workflow

3. 2. Data Collection

The dataset used in this study was sourced from the publicly available collection on Roboflow titled Oil Palm Detection by Manfred Michael with YOLOv12 format [16]. It consists of high-resolution aerial images of oil palm. The representative samples from the oil palm UAV imagery dataset can be seen in Fig 2. These images comprise a total of 4063 UAV images of oil palm trees, each with a resolution of 224×224 pixels with a .jpg extension. Annotated bounding boxes representing individual oil palm trees under different lighting and planting densities are also included in this dataset.



Fig. 2. Samples from the oil palm UAV imagery dataset

3. 3. Preprocessing and Data Distribution

Several preprocessing techniques are used to standardize and enhance the dataset before training to increase its robustness. Each image will be resized to 640×640 pixels to accommodate the model input from YOLO, while maintaining consistency and resolution in each oil palm tree image.

Additionally, each image will go through auto orientation correction, to automatically correct the direction (orientation) of the image before the image is processed further. To simulate lighting variations that are likely to occur on plantations in real time, the data augmentation process involves several steps, such as adding brightness between 0 and 15 percent of the original light, rotating the data slightly by about 15 degrees, and flipping the data horizontally to add perspective variations.

After all preprocessing has been applied, the next step is to divide the dataset into 70% for training, 20% for validation, and 10% for testing. Therefore, of the total 4,063 data sets, 2,844 images will be used for training, 813 for validation, and 406 for testing. This process aims to ensure that the data distribution process is carried out according to the correct composition and that the resulting model is able to predict the test data effectively.

3. 4. Model Training

The next step after preprocessing is model training. In this study, several nano versions of YOLO models were trained, such as YOLOv10n, YOLOv11n, and YOLOv12n, with hyperparameter tuning to compare their performance. The training process was carried out using the PyTorch deep learning framework and NVIDIA Quadro P6000 as a GPU for a robust computational data processing.

Each model was trained with 50 epochs, in accordance with previous research. Each YOLO was also trained using different batch sizes, which are: 8, 16, 32, 64 to compare the effectiveness and stability of each model. By varying the

batch size, we can assess the model's memory usage, gradient estimation performance, and generalizability.

The Adam optimizer is used to train each model using a few defined variables, such as weight decay = 0.0005, momentum = 0.937, final learning rate factor = 0.01, and initial learning rate = 0.002. Additionally, this research employed default data augmentation values with mosaic augmentation enabled, such as HSV (hue = 0.015, saturation = 0.7, value = 0.4), random scaling = 0.5, translation = 0.1, and horizontal flipping = 0.5. To ensure that changes in values are primarily due to differences in batch size and model design and not by other variables, this configuration makes sure that each variation is equalized.

On the other hand, the YOLO method utilizes the features of the nano version, which is the lightest form and was selected especially for quick inference with minimal processing costs while maintaining excellent accuracy. This served as the foundation for determining the YOLO model for drone images. At the time of the research, the most recent models that were accessible to the public were YOLOv10, YOLOv11, and YOLOv12. This work is anticipated to have an impact on identifying the optimal algorithm that can be applied to drone images, particularly in oil palm plantation environments, by comparing the three YOLO models with different batch sizes.

3. 5. Evaluation

The next stage is to evaluate the trained models' performance after obtaining many models with varying batch sizes. For easier comprehension, each model will be mapped to a table. The objective of this efficient measurement is to identify oil palm plants using drone-captured aerial imagery with the best computational efficiency and accuracy. The measuring metrics utilized for object detection include:

- Precision: The true positive ratio, or the number of bounding boxes accurately predicted out of all predicted bounding boxes. A higher precision number means a lower false positive rate.

$$Precision = \frac{TP}{TP+FP} \quad (1)$$

- Recall: The proportion of true positives among all actual positives in the ground truth. High recall indicates that the most relevant objects are discovered.

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

- mean Average Precision at IoU 0.5 (mAP@0.5): Predictions with Intersection over Union (IoU) larger than 0.5 have a higher average accuracy over ground truth across all classes. This measure achieves a balance between precision and recall at a standard threshold.
- mean Average Precision at IoU 0.5 to 0.95 (mAP@0.5:0.95): The mAP average was calculated in 0.05-step increments between thresholds 0.5 and 0.95. It offers a more comprehensive evaluation of the quality of detection.
- Latency: Latency describes the average time required to process an image during testing, measured in milliseconds (ms). This measurement is particularly important for applications that will be used in real time

- **GPU Memory Usage:** GPU Memory Usage describes the amount of memory used during the inference and training process, which is very helpful for considering computing resource requirements.

The best model among these six measurement variables will be found after a training procedure using both training and validation data. Testing will then continue using 406 images from the test dataset that have never been added to the model before to monitor real-world situations. Each step in the suggested research framework can offer a comparison to the YOLO version model that can be used to identify oil palm trees using images from drones in plantation environments in order to determine the best configuration that maximizes

accuracy and computational efficiency in oil palm tree detection.

IV. EXPERIMENTAL RESULTS

The YOLOv10n, YOLOv11n, and YOLOv12n models trained on the oil palm detection dataset for different batch sizes (8, 16, 32, 64) are thoroughly evaluated in this section. Precision, recall, mean Average Precision at IoU 0.5 (mAP@50), and mean Average Precision averaged across IoU thresholds 0.5 to 0.95 (mAP@50-95) are the main detection measures evaluated. Furthermore, there are measurements for computational efficiency including inference latency and GPU memory allocation. The performance metrics for different YOLO model combinations and batch sizes are shown in Table 1.

TABLE I. PERFORMANCE METRICS FOR THE COMBINATIONS OF YOLO MODELS AND BATCH SIZE

YOLO Model	Batch Size	Precision	Recall	mAP@50	mAP@50-95	Latency (ms)	GPU Memory Usage (MB)
YOLOv10n	8	0.8598	0.8882	0.9008	0.5631	16.04	125.02
	16	0.8449	0.8811	0.9052	0.5629	16.25	133.78
	32	0.8720	0.8730	0.8918	0.5605	16.54	81.52
	64	0.8604	0.8791	0.8944	0.5583	16.52	123.42
YOLOv11n	8	0.8465	0.8857	0.8956	0.5560	17.49	51.95
	16	0.8505	0.8843	0.8979	0.5607	16.28	134.77
	32	0.8552	0.8811	0.8967	0.5671	16.42	41.97
	64	0.8447	0.8923	0.8996	0.5619	15.70	114.38
YOLOv12n	8	0.8501	0.8993	0.8990	0.5613	22.13	125.60
	16	0.8635	0.8875	0.9022	0.5643	22.42	124.65
	32	0.8746	0.8650	0.8945	0.5557	20.76	124.82
	64	0.8572	0.8892	0.8973	0.5618	20.85	123.64

Table 1 shows that recall rates are over 89% and precision values are generally above 85%. This clearly shows that all three YOLO versions (YOLOv10n, YOLOv11n, and YOLOv12n) perform well in object detection, especially for oil palm trees. This experiment also indicates that there aren't many false positives or false negatives when using YOLO to identify oil palm trees. The model's ability to produce predictions with good overlap which can be seen at Intersection over Union (IoU) with the actual bounding boxes is demonstrated by the mAP@50 values, which are approximately 0.89 or higher across all tests. Furthermore, the models' detailed localization quality can be inferred from the mAP@50-95 metric, which shows the average performance across a range of stringent IoU thresholds (from 0.5 to 0.95). The models' ability to consistently place and size bounding boxes accurately is demonstrated by the values near 0.56. In terms of speed, YOLOv11n with a batch size of 64, stands out with the lowest latency of 15.70 ms per image, making it suitable for real-time oil palm trees monitoring using drones. The most efficient use of GPU memory is found with YOLOv11n with a batch size of 32, which only requires around 41.97 MB of allocated memory, showing that resource efficiency can be achieved without compromising detection performance. The sample detection results of oil palm trees with confidence score can be seen at Fig 3.



Fig. 3. Detection results of oil palm trees with confidence score

In the previous study by Prasvita et al. [10], the best-performing model for oil palm tree detection was YOLOv5s with a batch size of 32. The model achieved an mAP@50 of 0.851, an mAP@50-95 of 0.457, and an F1-score of 0.785, indicating solid performance on UAV imagery. In comparison, our experiments using more recent YOLO variants (YOLOv10, YOLOv11, and YOLOv12) achieved an average mAP@50 of 0.8979 and mAP@50-95 of 0.5611. These findings indicate that more recent YOLO architectures

offer better detection performance. It is crucial to remember that due to variations in datasets and experimental conditions, this comparison is not totally straightforward.

The three YOLO nano versions offer appealing alternatives that can be chosen based on application needs:

- For high speed and throughput, **YOLOv11n** with a batch size of **64** is highly recommended (Latency 15.70 ms per image).
- For memory efficiency while maintaining strong performance, **YOLOv11n** with a batch size of **32** is the ideal choice (requires only 41.97 MB and mAP@50-95 with highest overall accuracy 0.5671).

Although YOLOv12 is the latest iteration in the YOLO series, the experimental results reveal it does not outperform YOLOv11 in several key detection metrics for oil palm tree detection. The YOLOv12 showed promising results, particularly in terms of precision while recall was comparable, it was outperformed by YOLOv11 in both speed and memory efficiency. On the other hand, YOLOv10 delivered a competitive accuracy compared to YOLOv11. YOLOv10, on the other hand, needed more memory consumption and perform with more slowly during detection.

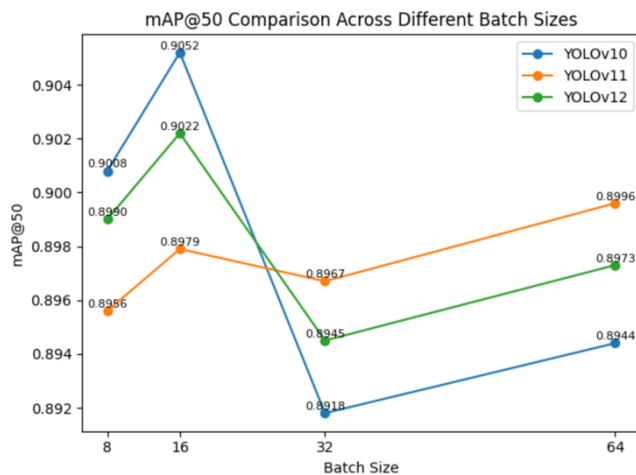


Fig. 4. mAP@50 Comparison Across Different Batch Sizes

Fig. 4 illustrates the comparison of mAP@50 across different batch sizes for YOLOv10, YOLOv11, and YOLOv12. Overall, all models show relatively stable performance with only minor variations. While YOLOv10 achieves the highest mAP@50 at batch size 16, YOLOv11 demonstrates more consistent performance across all batch sizes. This consistency, combined with its efficiency in terms of latency, makes YOLOv11 a more suitable choice for real-time oil palm detection in drone-based applications.

V. CONCLUSIONS

This study examined the use of the most advanced YOLO nano models—YOLOv10n, YOLOv11n, and YOLOv12n—for aerial imagery-based oil palm tree detection. The models were thoroughly pre-processed and trained on a publicly available dataset to increase detection robustness. According to the experiment, all three YOLO models show remarkable performance, with mean Average Precision (mAP) of approximately 0.89 for IoU threshold 0.5 and precision and recall above 85%. The fastest inference speed (15.70 ms per image) was consistently attained by YOLOv11n at batch size 64, which makes it ideal for real-time monitoring applications. With its highest mAP@50-95 (0.5671) and

remarkably low GPU memory allocation (41.97 MB), YOLOv11n at batch size 32 stood out as an example of efficiency in resource-constrained settings. Despite achieving the highest recall and precision levels, YOLOv12n may not be as suitable for applications requiring fast calculation because to its greater latency. These findings suggest that YOLOv11n, which provides a suitable balance between accuracy, speed, and computational resource consumption, is the best model for oil palm detection tasks in drone-based precision agriculture. The most recent YOLO nano models are successfully integrated with practical applications in oil palm plantation monitoring in this study. However, experimental validation is limited using a single public dataset and the absence of implementation in real plantation situations. These constraints may limit how widely the findings can be used.

Images from a range of locations and environments should be included in future work. Furthermore, it will be important to address deployment issues like field computing limitations and integration with mobile drone systems. Additionally, some statistical tests would be added to strengthen the experiment. Furthermore, this research can also be expanded by focusing on smallholder oil palm plantations and industrial oil palm plantations, so that a comparison can be made regarding the accuracy of oil palm plantations. The purpose of this paper is to provide the foundations for future research into automated, real-time monitoring systems that can improve the sustainability and efficiency of plantation management by highlighting these research directions and challenges.

ACKNOWLEDGMENT

Data Availability

The data used in this study was taken from Roboflow which can be accessed open source at the link: <https://universe.roboflow.com/manfred-michael/oil-palm-detection>.

Credit Authorship Contributor Statement

Jeklin Harefa: Dataset Gathering, Modelling, Experiment, Writing – Original Draft; **Haryono Soeparno:** Supervision, Writing – Review & Editing; **Yulyani Arifin:** Supervision, Writing – Review & Editing; **Ford Lumban Gaol:** Supervision, Writing – Review & Editing.

AI Usage Declaration

This research utilized Large Language Models tools, such as ChatGPT and Turnitin, to assist in the paper's reference writing. To preserve the integrity of the research content, all recommendations and input from the LLM tools were examined and modified.

REFERENCES

- [1] A. D. Santoso, E. Sribudiani, A. Priyanti, D. Yulistiani, H. Daulay, R. Hanifa, I. G. A. P. Mahendri, Priyono, U. K. Yaumidin, A. Ali, E. Erwan, D. Iskandar and I. N. Djarot, "Sustainable utilization of palm oil industry by-products for livestock feed: A digestibility and environmental assessment," *Case Studies in Chemical and Environmental Engineering*, vol. 12, 2025.

- [2] S. S. Lee, L. G. Lim, S. Palaiahnakote, J. X. Cheong, S. S. M. Lock and M. N. B. Ayub, "Oil palm tree detection in UAV imagery using an enhanced RetinaNet," *Computers and Electronics in Agriculture*, vol. 227, 2024.
- [3] S. A. Hashim, S. Daliman, I. N. M. Rodi, N. A. Aziz, N. A. Amaludin and A. E. Rak, "Analysis of Oil Palm Tree Recognition using Drone-Based Remote Sensing Images," in *IOP Conference Series: Earth and Environmental Science*, 2020.
- [4] J. Redmon and A. Farhadi, "YOLO9000: Better, Faster, Stronger," in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, USA, 2017.
- [5] M. M. Billah, A. A. Rakib, M. I. Haque, A. S. Ahamed, M. S. Hossain and M. K. N. Borsha, "Real-Time Object Detection in Medical Imaging Using YOLO Models for Kidney Stone Detection," *European Journal of Computer Science and Information Technology*, vol. 12, no. 7, pp. 54-65, 2024.
- [6] Z. Su, J. Yu, H. Tan, X. Wan and K. Qi, "MSA-YOLO: A Remote Sensing Object Detection Model Based on Multi-Scale Strip Attention," *Sensors*, vol. 23, no. 15, p. 6811, 2023.
- [7] J. Tang, C. Ye, X. Zhou and L. Xu, "YOLO-Fusion and Internet of Things: Advancing object detection in smart transportation," *Alexandria Engineering Journal*, vol. 107, pp. 1-12, 2024.
- [8] C. M. Badgujar, A. Poulose and H. Gan, "Agricultural object detection with You Only Look Once (YOLO) Algorithm: A bibliometric and systematic literature review," *Computers and Electronics in Agriculture*, 2024.
- [9] J. Josdaan, V. C. Tamsil, J. Harefa and K. Jingga, "Revolutionizing palm oil ripeness classification: Utilizing YOLOv8 for ultra-precise ripeness detection," in *9th International Conference on Computer Science and Computational Intelligence 2024 (ICCSCI 2024)*, 2024.
- [10] D. S. Prasvita, A. M. Arymurthy and D. Chahyati, "Deep Learning Model for Automatic Detection of Oil Palm Trees in Indonesia with YOLO-V5," in *SIET '23: Proceedings of the 8th International Conference on Sustainable Information Engineering and Technology*, Badung, Bali, Indonesia, 2023.
- [11] Y. A. A. Soetrisno, E. Handoyo, Sumardi and E. W. Sinuraya, "OIL PALM LEVEL OF RIPENESS CLASSIFICATION USING EFFICIENTDET-LITE CNN ARCHITECTURE," *Journal of Oil Palm Research*, pp. 618-629, 2024.
- [12] A. Minarto, M. H. Ramadhan, A. Lie and E. Irwansyah, "Oil palm tree counting and abnormality assessment using deep learning and index vegetation," in *9th International Conference on Computer Science and Computational Intelligence 2024 (ICCSCI 2024)*, 2024.
- [13] M. G. Naftali, G. Hugo and Suharjito, "Palm Oil Counter: State-of-the-Art Deep Learning Models for Detection and Counting in Plantations," in *IEEE Access*, 2024.
- [14] A. Haq, E. S. Cahyadi, M. S. Kasyfillah, R. Sarno and A. T. Haryono, "Multi-class Oil Palm Trees Condition Detection from UAV Images using Faster R-CNN with EfficientNetV2," in *2024 2nd International Conference on Technology Innovation and Its Applications (ICTIIA)*, Medan, Indonesia, 2024.
- [15] K. H. Heng, A. A. Malik, M. A. B. Osman, Y. Yusop and I. H. Hamzah, "Healthy and Unhealthy Oil Palm Tree Detection Using Deep Learning Method," *IJACSA International Journal of Advanced Computer Science and Applications*, pp. 748-757, 2025.
- [16] M. Michael, "Oil Palm Detection Dataset," June 2022. [Online]. Available: <https://universe.roboflow.com/manfred-michael/oil-palm-detection>.